

TradeBot

Complete User Guide

14 Built-in Strategies

9 stock + 4 crypto + EMA + Patterns

10-Layer Risk Engine

Kill switch, loss limits PDT, exposure caps

Stocks & Crypto 24/7

Alpaca, Tradier, Binance spot

AI-Powered

Explanations + weekly auto-tuner

Backtesting Studio

Historical simulation + drift monitor

Webhooks & Alerts

TradingView signals
Slack/Discord/Telegram

Automated algorithmic trading · 60-second engine tick · Multi-layer risk controls · All from one dashboard

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What is TradeBot?

TradeBot is an automated algorithmic trading platform that connects to your broker accounts, runs 14 built-in strategies on a 60-second engine tick, enforces multi-layer risk controls, explains every trade with AI, and supports both stocks and crypto — all from one web dashboard.

It runs continuously on your PC or a VPS. No coding required.

Key Principles

- **Paper-first** — defaults to paper trading. No real money at risk until you explicitly add a live account.
- **Transparent** — every signal, block, and order is logged with a full reason.
- **Risk-first** — 10 configurable guards run before every order.
- **AI-assisted** — every trade gets a plain-English explanation and weekly parameter tuning.

WARNING

TradeBot is a tool, not financial advice. Past strategy performance does not guarantee future results.
Always start with paper trading.

Installation & First-time Setup

System Requirements

- Windows 10/11 64-bit or Ubuntu 22.04/24.04 LTS (VPS)
- Python 3.11 or newer
- Internet connection · 500 MB free disk space

Windows — Quick Start

1 Run setup.bat

Double-click setup.bat. Installs Python if missing, creates a venv, and installs all pip dependencies.

2 Configure .env

Create a .env file in the project folder. Set DB_SECRET_KEY — generate with: `python -c "from cryptography.fernet import Fernet; print(Fernet.generate_key().decode())"` — back this up securely.

3 Start the server

Double-click start.bat or run: `uvicorn server.main:app --port 8000`. Dashboard opens at `http://localhost:8000`

4 Set your password

On first launch you are prompted to create a dashboard password.

5 Add a broker account

Broker Accounts → Add Account → pick broker → enter API keys → Test.

6 Enable a strategy

Bots & Strategies → select account → toggle a strategy on.

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Keep your .env file private. It contains your AES-256 encryption key. Never commit it to version control.

Broker Accounts

TradeBot supports multiple broker accounts simultaneously. Each has its own Paper/Live type, credentials, and strategy assignments — managed independently.

Supported Brokers

Broker	Asset Class	Paper Trading
Alpaca	US Stocks & ETFs	Separate paper API keys (start with PK)
Tradier	US Stocks & Options	Sandbox API at sandbox.tradier.com
Binance	Crypto Spot (USDT pairs)	Demo keys from demo.binance.com

Adding an Account

1 Click Add Account

Broker Accounts page → Add Account → choose broker from the tabbed picker (Stocks, Crypto, Forex, Futures).

2 Select Paper or Live

Each account is independently Paper or Live. Binance paper requires demo keys from demo.binance.com — separate from live keys.

3 Enter API credentials

Paste your API Key and Secret. Encrypted with AES-256 immediately. The secret is never stored in plaintext.

4 Test connectivity

Click Test on the account card. Green tick = credentials valid.

5 Assign strategies

Go to Bots & Strategies to link strategies to this account.

WARNING

If you lose `DB_SECRET_KEY` from `.env`, stored broker credentials cannot be decrypted. Back up `.env` securely.

Strategies & Bots

TradeBot ships with 14 automated strategies. The engine evaluates all enabled strategies every 60 seconds. Stock strategies only run during US market hours; crypto strategies run 24/7.

4.1 Stock Strategies (9 total)

Strategy	Signal Logic	Best For
Momentum	Top actives + price above 20-day SMA + RSI 40–70	Trending markets
RSI Mean Reversion	Buy RSI < 30; exit RSI > 50	Range-bound stocks
MACD + Volume	Bullish MACD crossover + above-average volume	Momentum with confirmation
Bollinger Bands	Buy lower band + oversold RSI; exit middle band	Low-volatility stocks
Golden Cross	50-day SMA crosses above 200-day SMA	Long-term trend following
SMA Crossover	Fast SMA crosses above slow SMA (configurable periods)	Short-term trend detection
52-Week Breakout	Price closes above 52-week high on strong volume	Breakout & momentum
EMA Confluence	8/21/50/200 EMAs all aligned; scaled sizing by count	High-conviction entries
Classic Patterns	Candlestick & continuation patterns; EMA-200 trend filter	Pattern-based swing trades

4.2 Crypto Strategies (4 total)

Strategy	Signal Logic
Crypto Trend	EMA crossover (fast vs. slow). Buy bullish, sell bearish.
Crypto RSI Bounce	Buy RSI < oversold threshold; sell RSI > overbought threshold.
Crypto Volatility Breakout	Buy lower Bollinger Band + oversold RSI; exit at SMA (middle band).
Crypto Grid	Divides price range into bands. Buys near band bottoms, sells near tops. Auto-detects range from recent history. Trend SMA filter prevents entries in downtrends.

Enabling a Strategy

1 Open Bots & Strategies

Select a broker account on the left. Compatible strategies appear on the right.

2 Toggle on

Flip the toggle. Picked up on the next 60-second tick.

3 Adjust parameters

Expand a strategy to edit notional size, indicator periods, symbol lists. Changes take effect immediately.

INFO

Stock strategies only run during US market hours (9:30 AM – 4:00 PM ET). Crypto runs 24/7.

Crypto Trading with Binance

TradeBot connects to Binance for 24/7 spot trading. All pairs trade against USDT (e.g., BTC/USDT, ETH/USDT). Paper and live accounts use completely separate API keys.

Getting Binance API Keys

1 Paper Trading — Binance Demo

Go to demo.binance.com, create an account, and generate API keys under API Management. Demo keys only work with the demo exchange.

2 Live Trading — Binance.com

Go to binance.com → API Management. Enable Spot & Margin Trading. Disable withdrawals for safety.

3 Add in TradeBot

Broker Accounts → Add Account → Crypto tab → Binance → select Paper or Live → paste keys → Test.

WARNING

Demo keys and live keys are completely separate. Using a live key on a Paper account (or vice versa) will cause an authentication error.

Common Binance Issues

- **DB_SECRET_KEY missing** — Without it the app cannot decrypt stored API keys and Binance ticks fail silently. Check your .env file.
- **Key mismatch** — Paper accounts need keys from demo.binance.com; live accounts need keys from binance.com.
- **Insufficient balance** — Minimum order is ~\$10 USDT. Orders below this are skipped with a log entry.
- **Neutral zone** — Crypto Grid only signals near band tops/bottoms. Price in the middle of a band = no signal. This is correct behaviour.

Manual Orders

Place trades directly from the Positions & Orders page without waiting for a strategy signal.

1 Select an account

Use the account dropdown inside the order ticket.

2 Choose Buy or Sell

Enter a ticker (e.g., AAPL or BTC/USDT) and click Quote for the live price.

3 Set amount

Switch between Shares and Dollars mode, enter your amount, then click Place Order.

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Manual orders bypass the strategy engine and all risk guards. They execute immediately as market orders.

Risk Controls

Every signal — from strategies or webhooks — passes through 10 configurable risk guards before an order is placed. The first failing guard blocks the trade and logs the reason.

Guard	What It Does
Kill Switch	Master emergency stop. All automated trading halts instantly.
Daily Loss Limit	Halts trading when today's P&L; drops below N% of starting equity.
Weekly Loss Limit	Same as daily but over a rolling 7-day window.
Consecutive Losses	Blocks new trades after N losses in a row.
PDT Protection	Prevents a 4th day-trade in a 5-day rolling window (accounts under \$25K, stocks only).
Max Positions	Caps total open positions — separate limits for stocks and crypto.
Symbol Exposure	Prevents allocating more than N% of portfolio to one symbol.
Max Orders/Day	Limits total orders submitted in one trading day.
Trading Hours	Only allows orders between configured start/end times ET. Crypto strategies are exempt.
No-Trade List	Blacklisted symbols are never traded regardless of signals.

TIP Recommended starting settings: max daily loss 2%, max positions 5, trading hours 9:30–15:45 ET, PDT protection on.

Kill Switch

The kill switch is your emergency stop. One click halts all automated trading instantly. Manual orders still work.

Global vs. Per-Account

The **global kill switch** on the Risk page stops all accounts at once. Each broker account card also has its own **per-account kill switch** — useful when you want to pause one account while others continue trading.

How to Activate

Risk page → red **Activate Kill Switch** button. A red banner appears across all pages. Click **Deactivate** to resume.

INFO The kill switch does NOT close existing positions. It only stops new orders. To close all positions, use the close-all button on the Positions page.

Take-Profit

Automatically close positions when they reach a target gain percentage, without needing a sell signal from the strategy.

How It Works

After every 60-second engine tick, TradeBot scans all open positions. Any position whose unrealized gain meets or exceeds the take-profit threshold triggers an immediate market sell. Stock and crypto accounts each have their own configurable take-profit percentage.

Configuration

Go to **Risk** → Take-Profit section. Set **Stock Take-Profit %** and **Crypto Take-Profit %** separately. Set to 0 to disable.

INFO Take-profit exits run even when the engine kill switch is active — they're designed to lock in gains regardless.

Price Alerts

Set price thresholds on any symbol. When price crosses your target, you receive an instant notification via all configured channels.

1 Go to Settings → Price Alerts

Enter a symbol, choose Above or Below, and set your target price.

2 Save the alert

Active alerts are checked every 60-second engine tick.

3 Get notified

When triggered, alerts fire to all configured channels (Slack, Discord, Telegram, email). The alert is marked triggered and won't re-fire.

Performance Analytics

Track trading history, per-strategy stats, per-account P&L, and equity curve over time.

Widget	Description
Equity Curve	Portfolio value over time — 1W, 1M, 3M, 1Y views.
Strategy Stats	Per-strategy: signals, fills, blocks, win rate, Sharpe ratio.
Attribution by Account	Breakdown of signals and fills per strategy × account.
Top Symbols	Most-traded symbols by signal count and fill rate.
Daily Activity	Bar chart of signals per day over the last 30 days.

Interpreting Fill Rate

Fill rate = filled signals ÷ total signals. Below 50% usually means risk guards are blocking most signals. Filter Logs to **Blocked** and read the reason column.

Strategy Health & Drift Monitor

Strategy Health compares live trading performance against backtest benchmarks. It detects when a strategy drifts from expected behaviour.

Status	Meaning	Recommended Action
No Drift	Live win rate within normal range of backtest.	No action needed.
Minor Drift	Live performance below backtest by a noticeable margin.	Monitor closely. Consider re-tuning.
Major Drift	Live performance significantly underperforms backtest.	Pause the strategy and re-tune parameters.
No Benchmark	No backtest has been run for this strategy yet.	Run a backtest to enable drift monitoring.

INFO Run a backtest for every strategy you enable. This establishes the benchmark for drift detection and gives the AI Tuner a reference for weekly optimisation.

Backtesting Studio

Test any of the 14 strategies against historical price data before risking real money. Saved runs persist for comparison.

1 Choose strategy & symbols

Select from the strategy dropdown. Enter tickers separated by commas (e.g., AAPL, MSFT or BTC/USDT, ETH/USDT).

2 Set date range & costs

Set start/end dates (30 days minimum), initial capital, position size %, commission %, and slippage %.

3 Run and save

Click Run Backtest. Results show total return, win rate, max drawdown, Sharpe ratio, and a full trade log. Name the run to save it.

Key Metrics

Metric	Meaning
Total Return	Net profit as % of starting capital over the test period.
Win Rate	% of trades that closed with a profit.
Max Drawdown	Largest peak-to-trough decline. Over 20% is a warning sign.
Sharpe Ratio	Risk-adjusted return. Above 1.0 acceptable; above 2.0 excellent.
Avg Trade P&L;	Average dollar profit/loss per completed trade.

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Past performance does not predict future results. Backtests use daily bar data and assume unlimited liquidity.

AI Tuning & Trade Explanations

Trade Explanations

Every trade execution queues an AI explanation job. The AI reads the signal reason (indicator values, crossover details, pattern names) and writes a 2–3 sentence plain-English explanation visible in the Logs page.

Two providers supported:

- **Claude** (Anthropic API) — requires ANTHROPIC_API_KEY in .env
- **Ollama** (local, free) — requires Ollama running at localhost:11434

Weekly Auto-Tuner

Every Sunday at 11 PM ET, the auto-tuner analyses each strategy's live win rate vs. the backtest benchmark and proposes adjusted parameters (RSI thresholds, EMA periods, notional sizes) to improve performance. Results appear on the AI Tuning page.

INFO Trigger a manual tuning run at any time from the AI Tuning page. Suggestions can be applied with one click.

Webhook Signals

TradeBot can receive trade signals from TradingView, custom scripts, or any HTTP client. Incoming signals pass through the full 10-layer risk engine before execution.

1 Get your URL and token

Settings → Webhook section. Copy your unique URL and secret token.

2 Send a POST request

POST to `/api/webhook/signal` with header `X-Webhook-Token:` and a JSON body.

Signal JSON Format

Field	Type	Description
symbol	string	Ticker (e.g., AAPL or BTC/USDT)
side	string	buy or sell
qty	number	Shares or units (use this OR notional)
notional	number	Dollar amount to trade (use this OR qty)
account_id	integer	Optional — target a specific broker account

INFO Webhook signals pass through all risk guards — exactly the same as strategy signals.

Notifications

Get alerted on trades, risk blocks, price alerts, and daily summaries via email, Telegram, Slack, or Discord.

Email (Gmail / SMTP)

1 Generate a Gmail App Password

Google Account → Security → 2-Step Verification → App passwords → generate one named TradeBot.

2 Fill in SMTP settings

Host: smtp.gmail.com | Port: 587 | Username: your Gmail | Password: the App Password.

Telegram

1 Create a bot via BotFather

Message @BotFather in Telegram → /newbot → follow prompts → copy the token.

2 Get your Chat ID

Start a chat with your bot, then visit api.telegram.org/bot/getUpdates to find your chat_id. Paste both into Settings.

Slack & Discord

- **Slack:** Apps → Incoming Webhooks → Add to Slack → copy webhook URL.
- **Discord:** Server Settings → Integrations → Webhooks → New Webhook → copy URL.

Paste the URL into **Settings** → **Notifications**.

Going Live with Real Money

WARNING

Only switch to live trading after at least 2–4 weeks of successful paper trading with results you understand.

Pre-Live Checklist

- Ran paper trading for at least 2–4 weeks
- Understand which strategies are enabled and why
- Daily loss limit is set (2% recommended)
- Position sizes are appropriate for your account size
- Notifications configured — email or Telegram at minimum
- Kill switch accessible from your phone / tablet

Switching to Live

Broker Accounts → Add Account → pick broker → select **Live** → enter live API keys → Test → assign strategies.

- **Alpaca live:** requires identity verification and a linked bank account at alpaca.markets.
- **Binance live:** generate keys at binance.com → API Management with Spot & Margin Trading enabled and withdrawals disabled.

VPS Deployment (Ubuntu)

Run TradeBot on a VPS for 24/7 operation without keeping your PC on.

Recommended Hosting

Provider	Plan	Price	Notes
DigitalOcean	Basic Droplet 1GB	\$6/mo	Ubuntu 22.04 LTS
Hetzner	CX22 (2 vCPU, 4GB)	~€4/mo	Best price/performance

One-Command Deploy

The included `deploy.sh` script handles everything:

- 1 Clone the repo**
`git clone https://github.com/your-repo/TradeBot.git /opt/tradebot-src`
- 2 Run deploy**
`cd /opt/tradebot-src && sudo bash deploy.sh`
- 3 Configure .env**
`sudo nano /opt/tradebot/.env` — set `DB_SECRET_KEY` and notification tokens.
- 4 Start the bot**
`sudo systemctl start tradebot`
- 5 Monitor**
`sudo systemctl status tradebot journalctl -u tradebot -f` (live log stream)

NOTE Dashboard is at `http://:8000`. Use a strong password and consider nginx + HTTPS for production.

Troubleshooting & FAQ

Q: Strategy not placing orders

A: Check: (1) Strategy toggled on for a broker account. (2) A risk guard blocking it — filter Logs to Blocked. (3) Stock strategy: is the market open? (4) Crypto: DB_SECRET_KEY set in .env? (5) Does the symbol meet strategy criteria?

Q: Binance not trading

A: Most common: (1) DB_SECRET_KEY missing from .env — decrypt fails silently. (2) Demo vs. live key mismatch. (3) Balance below \$10 USDT minimum. (4) Price in neutral zone (Crypto Grid only signals near band edges).

Q: Very low fill rate — mostly Blocked signals

A: A risk guard is blocking most trades. Filter Logs to Blocked and read the reason column — it names exactly which guard triggered. Adjust in Risk Controls.

Q: PDT rule block

A: Account under \$25,000 has used 3 day-trades in a 5-day window. Block lifts automatically. Fund above \$25,000 to become PDT-exempt. PDT does not apply to crypto.

Q: Daily loss limit triggered

A: Kill switch activated automatically. Review positions, adjust the limit if needed, then click Deactivate Kill Switch to resume.

Q: API key / decryption error

A: DB_SECRET_KEY in .env is wrong or missing. If you changed it after saving keys, stored credentials are unrecoverable — re-enter them. Always back up .env.

Q: Email notifications not arriving

A: Check spam folder. Use a Gmail App Password (not your regular password). SMTP: smtp.gmail.com, port 587.

Q: Where is my data stored?

A: All data lives in trading.db (SQLite) in the project folder. Never delete or overwrite this file — it holds all credentials, settings, and history.

Q: Can I run TradeBot 24/7?

A: Yes — the server runs continuously. Stock strategies skip orders outside market hours. Crypto runs 24/7. Use a VPS or keep your machine on.

TIP

Still stuck? Check the server console output. The engine logs every skip, block, and error with a full reason.